CS & IT ENGINEERING

Introduction to C Programming

Lecture No.2

Pankaj Sir

TOPICS TO BE COVERED

01 printf() function

02 scanf() function

03 Data Types

a = 10 ✗

a

10

2096

Numbers without a decimal point

int a = 10 ; ✓

Numbers with a decimal point

float b = 12.78 ;

Data Types

1) Primitive
- (i) int
- (ii) float
- (iii) char
- (iv) void, bool & other

2) Derived
- (i) Arrays
- (ii) Pointers
- (iii) string

3) User defined
- (i) structure
- (ii) Union
- (iii) typedef
- (iv) enum

1.) Integer (int)
- short int
  - unsigned short int +12, 178
  - signed short int +12, -12, 0
- int
  - signed int
  - unsigned int
- long int
- long long int

2.) float
- float
- double
- long double

age = -ve X

3.) char
- unsigned
- signed

{To store 1 symbol}

Pankaj — 6 symbols

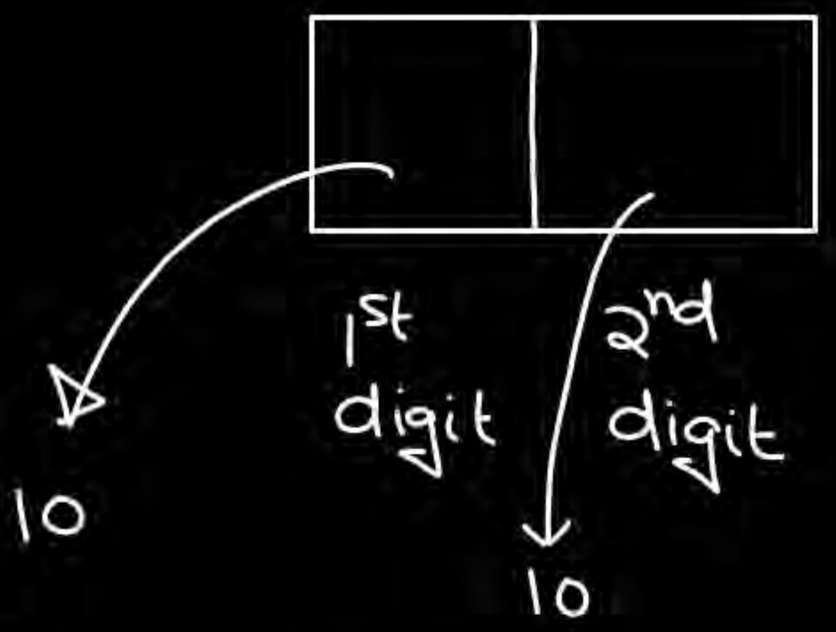
```
char ch = '@';
char ch = '#';
```

To store 1 symbol

decimal no. System (10 Symbols)

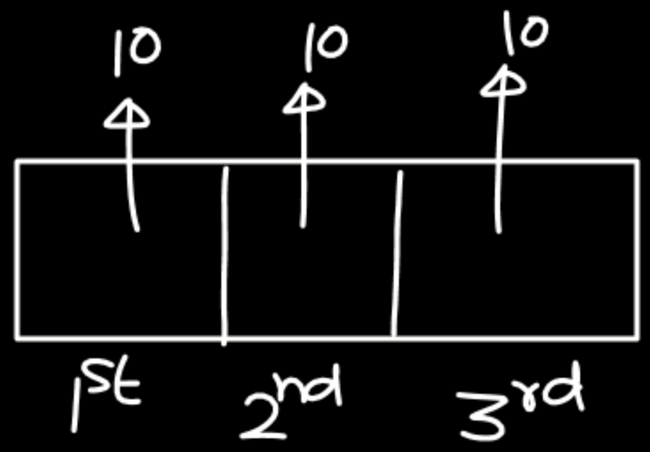
0, 1, 2, . . 9

Binary no. System (2 symbols — 0, 1)



0 - 9

possible values $\Rightarrow 10^2$

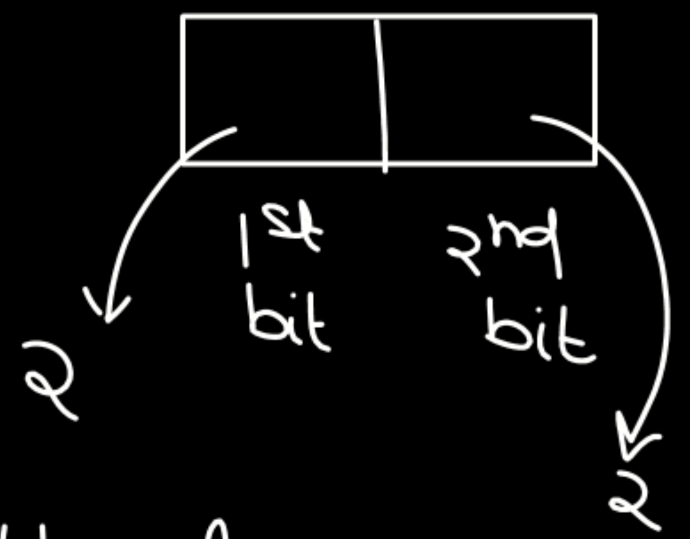


0 - 9

possible values $\Rightarrow 10^3 = 1000$

1 digit in a comp $\Rightarrow$ 1 bit

possible values $= 2^1$



0 0
0 1
1 0
1 1

possible values $= 2 \times 2 = 2^2$

$\longleftarrow$ 8 bits = 1 byte $\longrightarrow$

possible values = 2^8 = 256

4 values

00	$\rightarrow$ 0
01	$\rightarrow$ 1
10	$\rightarrow$ 2
11	$\rightarrow$ 3

Q 2 bytes = 16 bits

possible values $\Rightarrow$ 2^{16} (65536)

0 to 65535

Assuming short $\Rightarrow$ 2 byte

short int a = 10;

short int $\rightarrow$ unsigned short int

short int $\rightarrow$ signed short int

Unsigned short int : 2 bytes = $2 \times 8 = 16$ bits

possible values = 2^{16} (65536)

$\boxed{0 \text{ to } 2^{16}-1}$

signed short

-ve | +ve

32768 -ve values | 32768 +ve values

-32768 . . . . . , -2 , -1 | 0, 1, 2, . . . 32767

range : -32768 to 32767

2 byte ⇒ 16 bits

possible values = 2^{16} = 65536

-ve

$\frac{65536}{2}$ | $\frac{65536}{2}$

int $\Rightarrow$ 4 bytes

Unsigned int   4 bytes $\Rightarrow$ 32 bits

possible values $\Rightarrow 2^{32}$

0 to $2^{32}-1$

$$2^{10} = 1024 \approxeq 10^3$$

signed int $\Rightarrow$ 4 bytes $\Rightarrow$ 32 bits

$\dfrac{2^{32}}{2^{1}} \Rightarrow 2^{31}$

2^{32} possible values

2^{31} -ve values

2^{31} +ve values

$-2^{31} \ldots, -2, -1$

0 to $2^{31}-1$

range : -2^{31} to $2^{31}-1$

n bits

2^n possible values

unsigned

0 to $2^n - 1$

signed

2^{n-1} $\quad$ 2^{n-1}

$-2^{n-1}, \ldots, -2, -1,$ $\quad$ 0 to $2^{n-1} - 1$

Range : $\boxed{-2^{n-1} \text{ to } 2^{n-1} - 1}$

Short int a ; {signed}

short a ; {signed}

Signed short int a ;

Signed short a ;

Same

Unsigned short int b ;

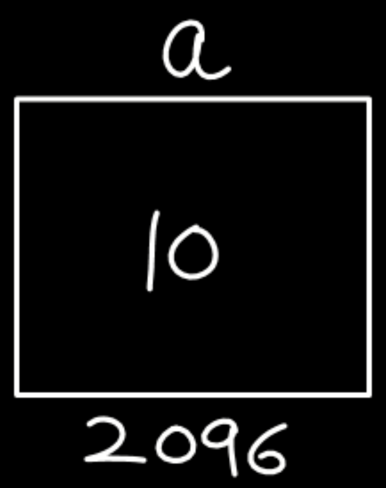
Unsigned short b ;

```c
#include<stdio.h>
void main(){

    int a = 10;

  printf("a");

      }
```



O/P :

```
a
```

printf ("10 + 10")

O/P : 10 + 10

format

" "

Textual data

format specifiers

specify format data

```c
#include<stdio.h>
void main(){

    int a = 10;

    printf("The value of a is %d", a)

}
```

format specifier for integer

a

10

The value of a is 10

%d → decimal number system

%d → short int , int

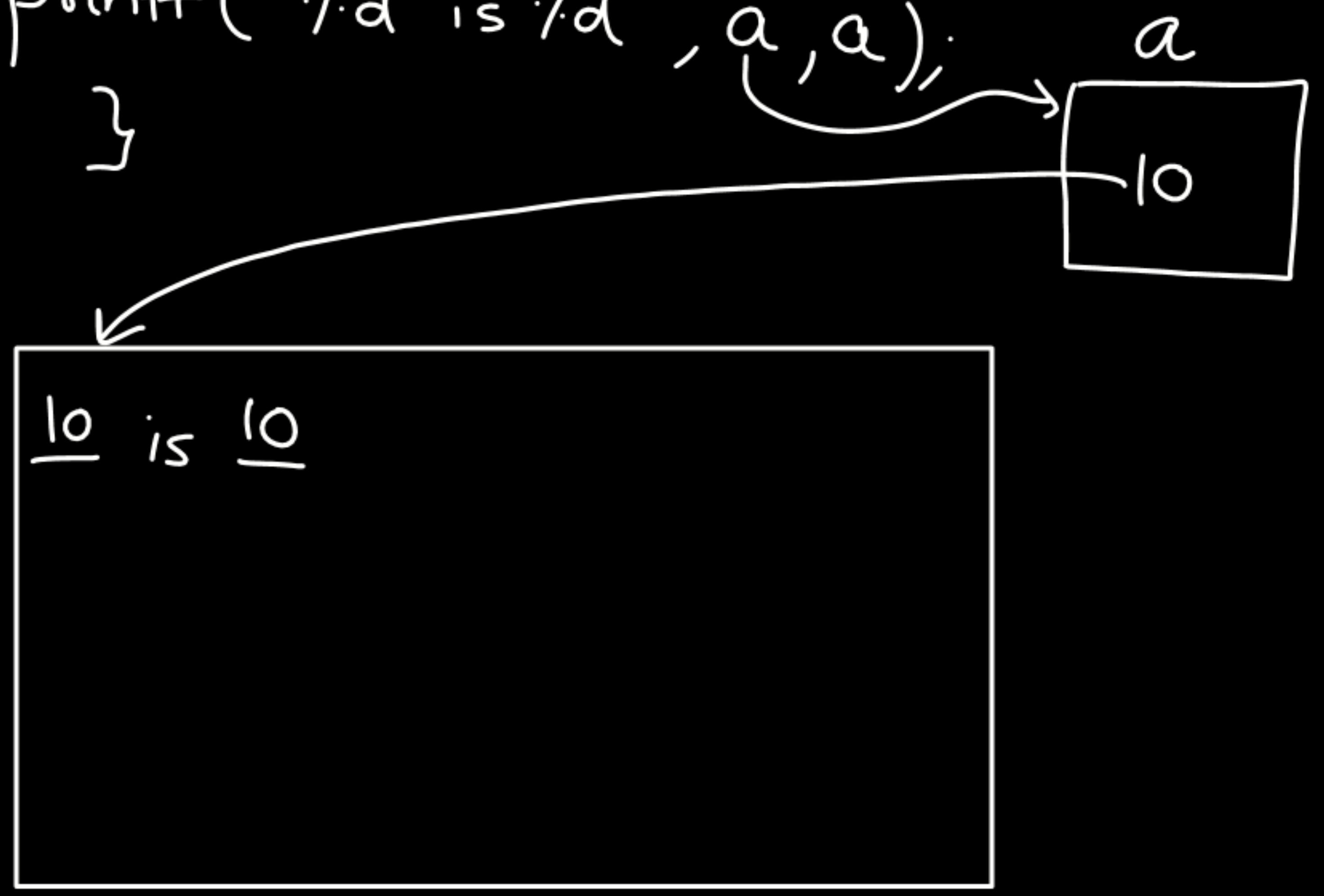
%ld → long int

%lld → long long int

%f → float

%c → char

```c
#include<stdio.h>
void main(){
        int a = 10;
        printf(" %d is %d ", a, a);
    }
```



a	b	c
10	20	30

```c
#include<stdio.h>
void main(){
    int a=10,b=20,c;
    c = a + b;

    printf("The sum of %d and %d is %d", a,b,c);
    printf ("The sum of %d and %d is %d", b,a,c);
}
```

```
The sum of 10 and 20 is 30
The sum of 20 and 10 is 30
```

char $\rightarrow$ 1 byte $\rightarrow$ 8 bits

2^8 possible values

unsigned char

0 to 2^8-1

0 to 255

Confusion

signed char

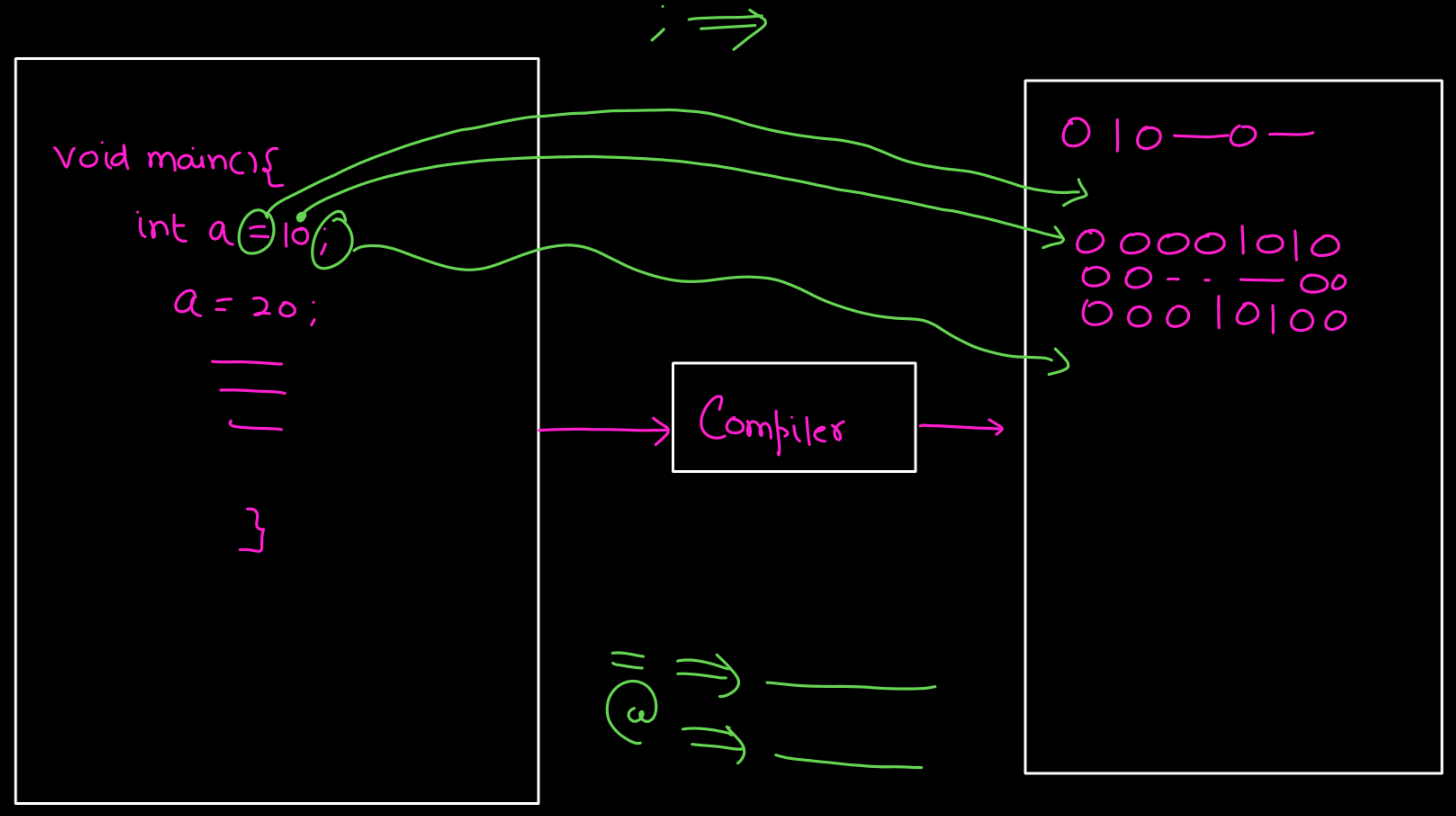
2^8 values

2^7 2^7

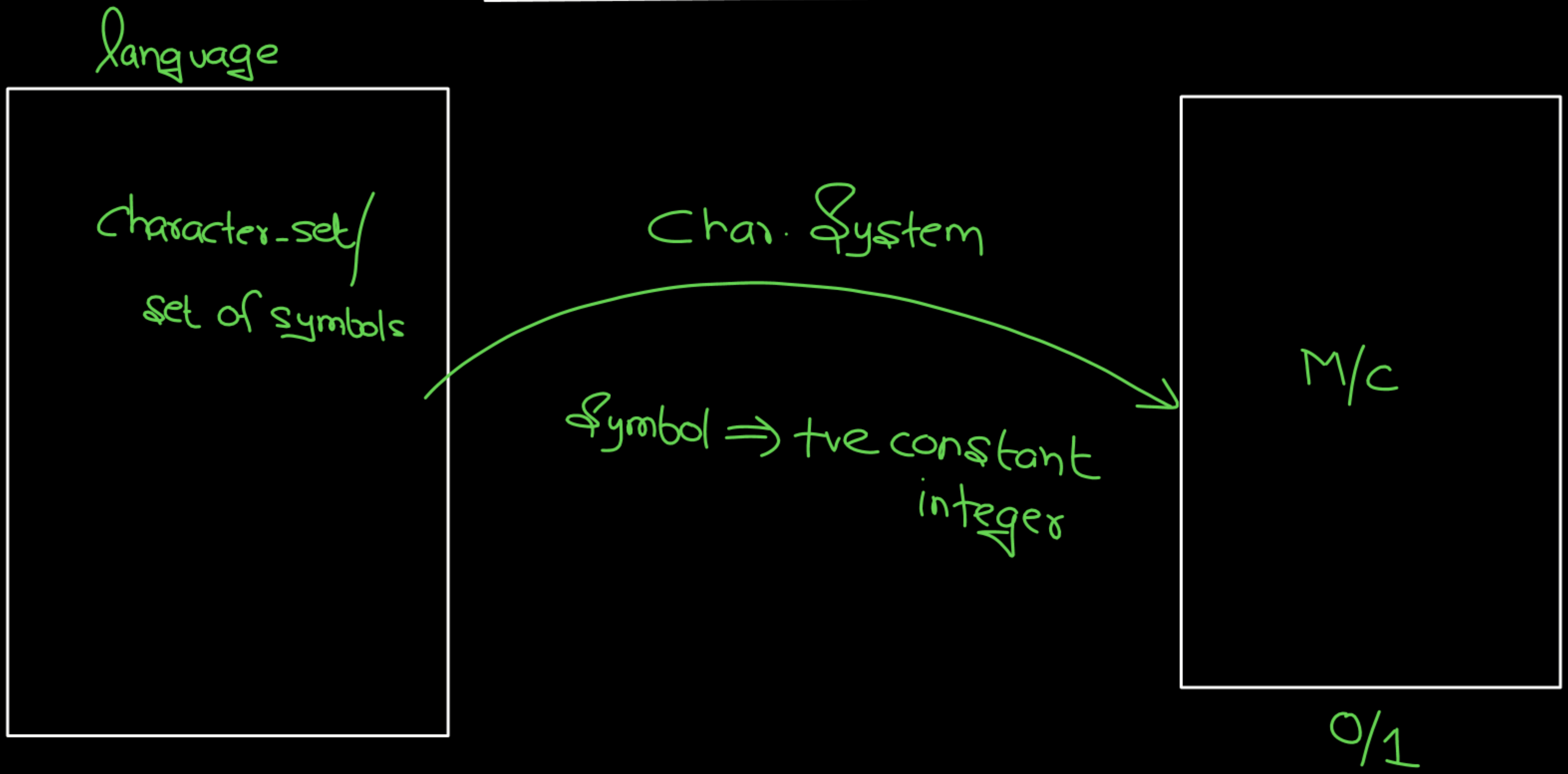
$-2^7 \ldots -1$ $0, 1, \ldots 2^7-1$

range : -2^7 to 2^7-1

: -128 to 127

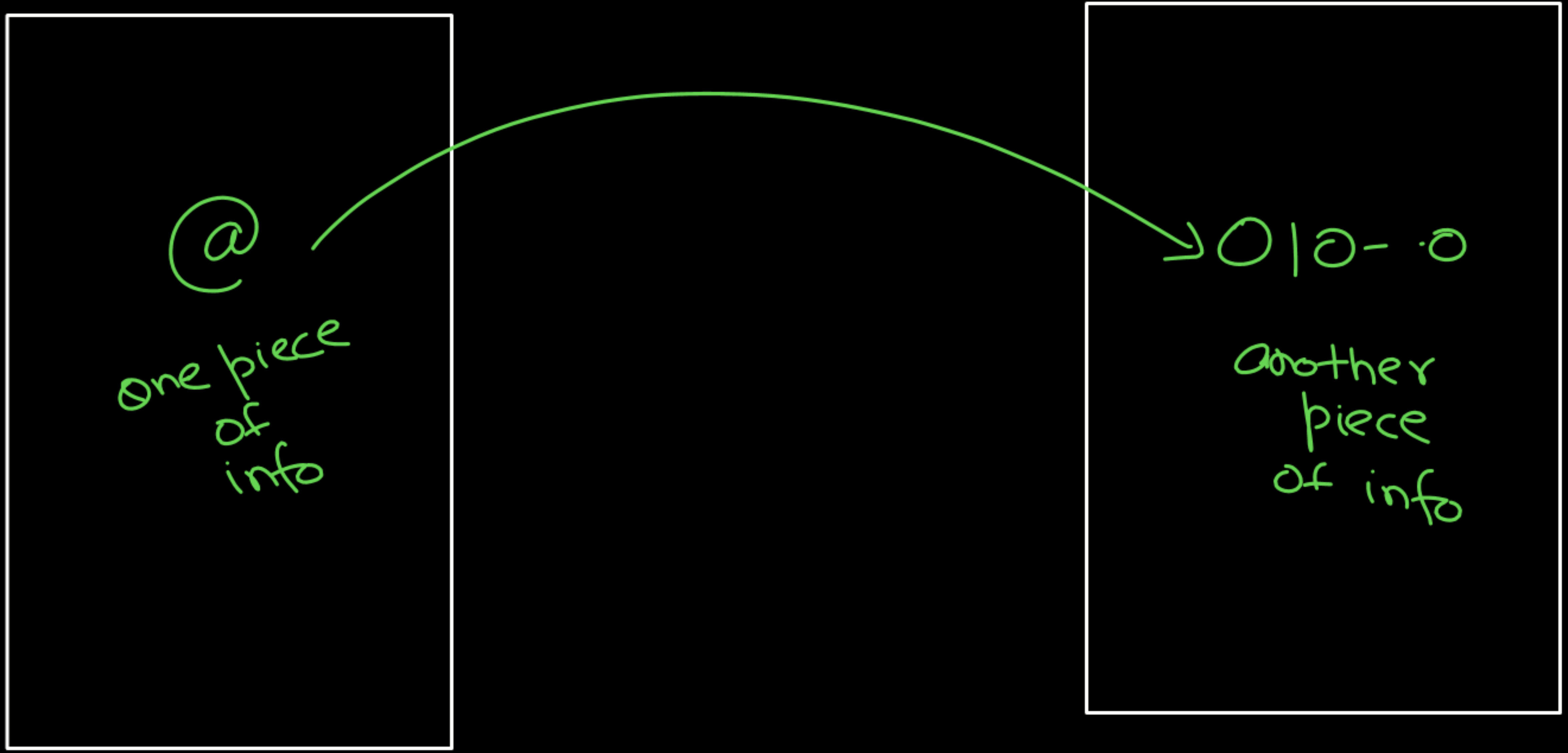

void main(){
int a = 10;
a = 20;
}
Compiler
0 1 0 — 0 —
00001010
00 - - — 00
00010100

CHARACTER SYSTEM



ASCII

American Standard Code for Information Interchange



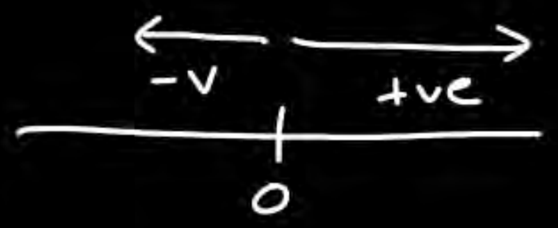
ASCII

A	65	a — 97	0 — 48
B	66	b — 98	
⋮		⋮	
Z	90	z — 122	

+ve value

Character set / Set of symbol

Cyclic Property

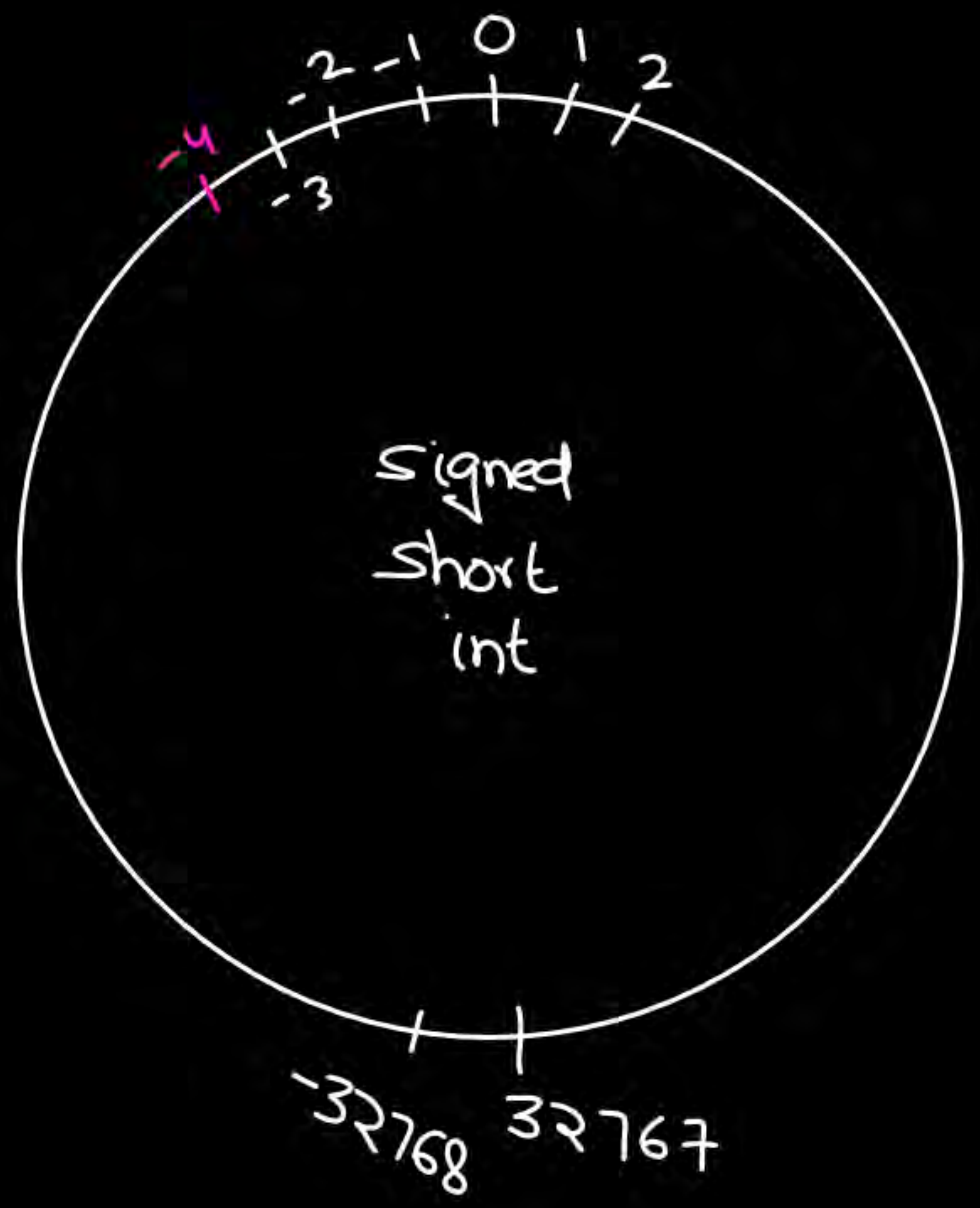


by default signed

short int a = -4;

-32768 to 32767

printf("%d",a); -4 ✓



-32768 to 32767

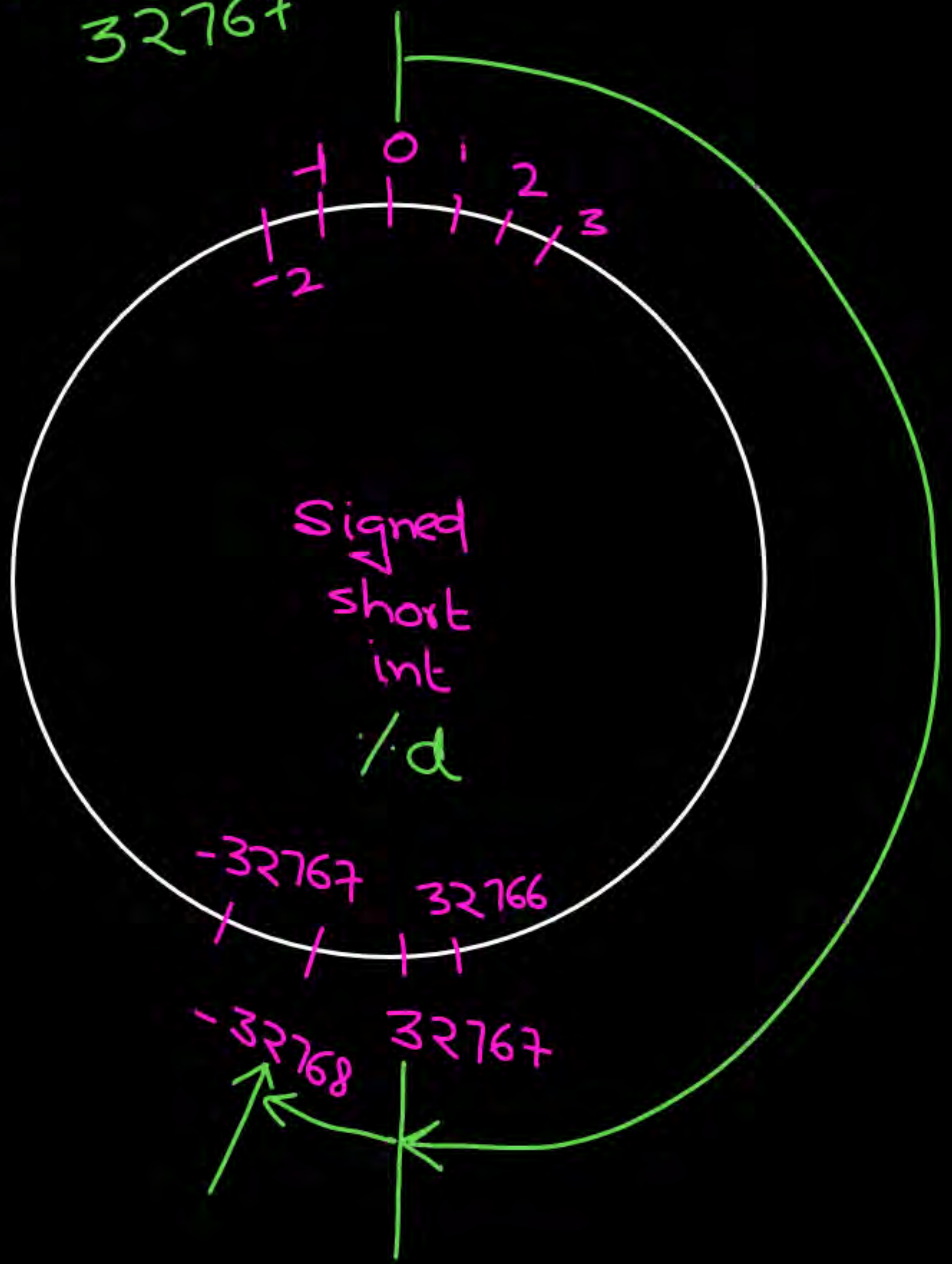
```c
short int a = 32768;
```
⇒ 1 more than 32767

```c
printf("%d", a);
```
-32768

assuming 2 bytes

a

-32768



-32768 to +32767

```
signed short i = -32770;
printf("%d", i);
```

2 less than -32768

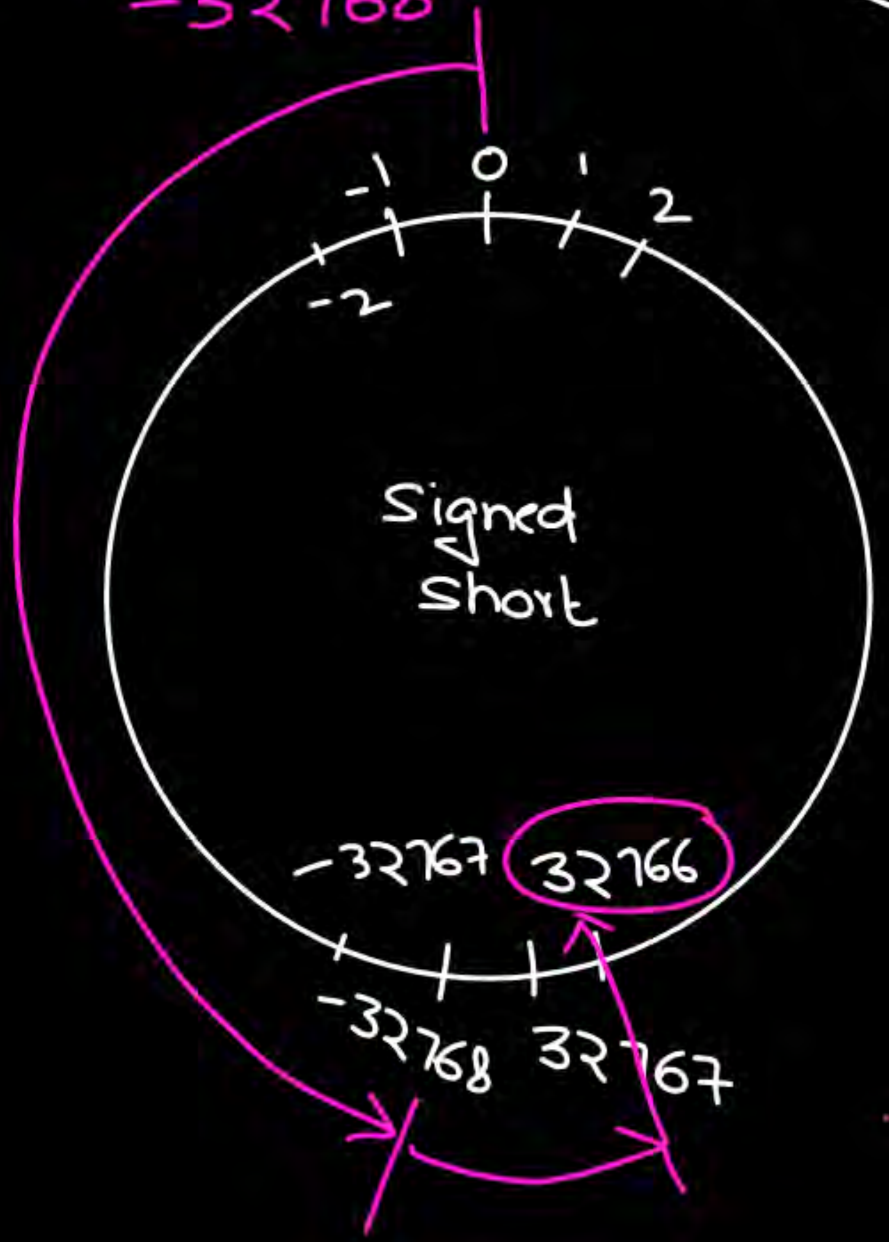
ASCII

Unicode

i

32766

32766

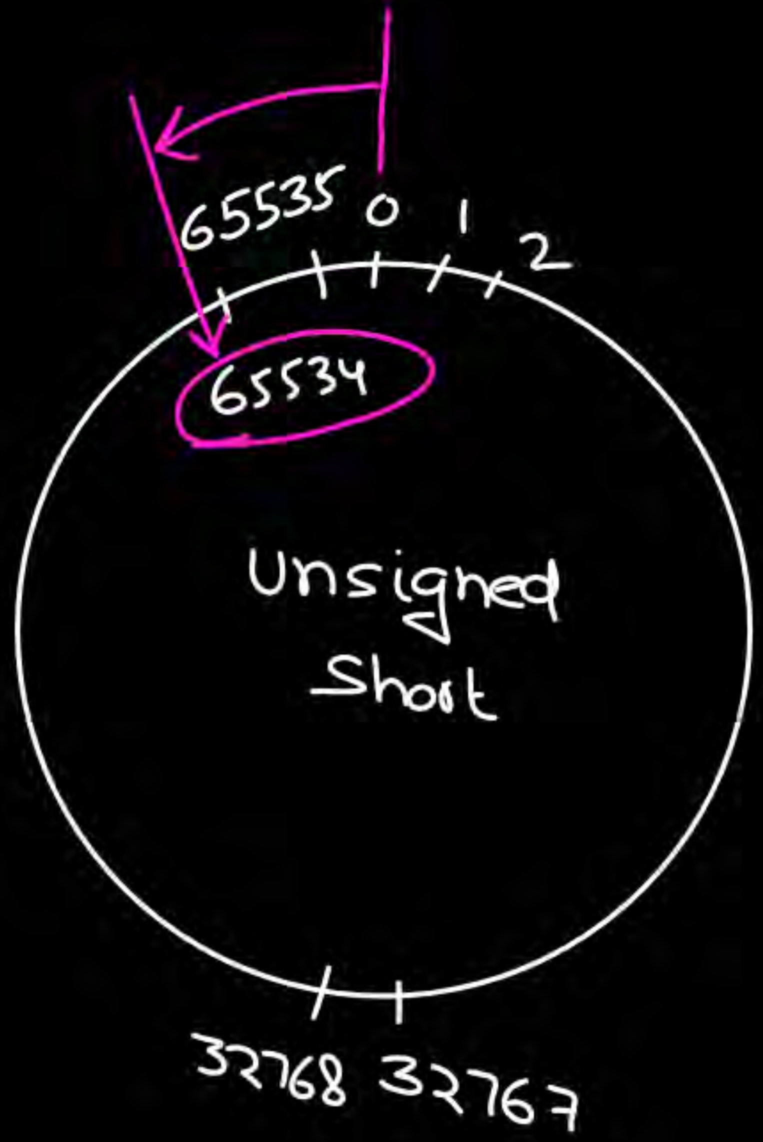


0 to 65535

```
unsigned short a = -2;
printf("%u",a);
```

65534

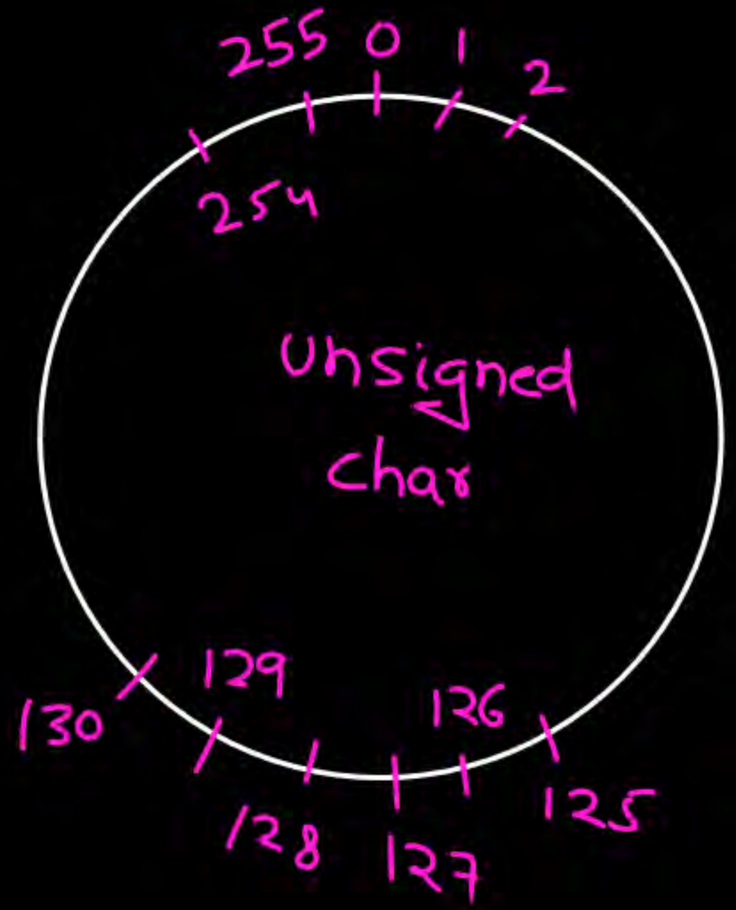
$\boxed{65534}$



```
char ch = 129 ; {signed}
printf("%d", ch); -127
```

ch

$\boxed{-127}$

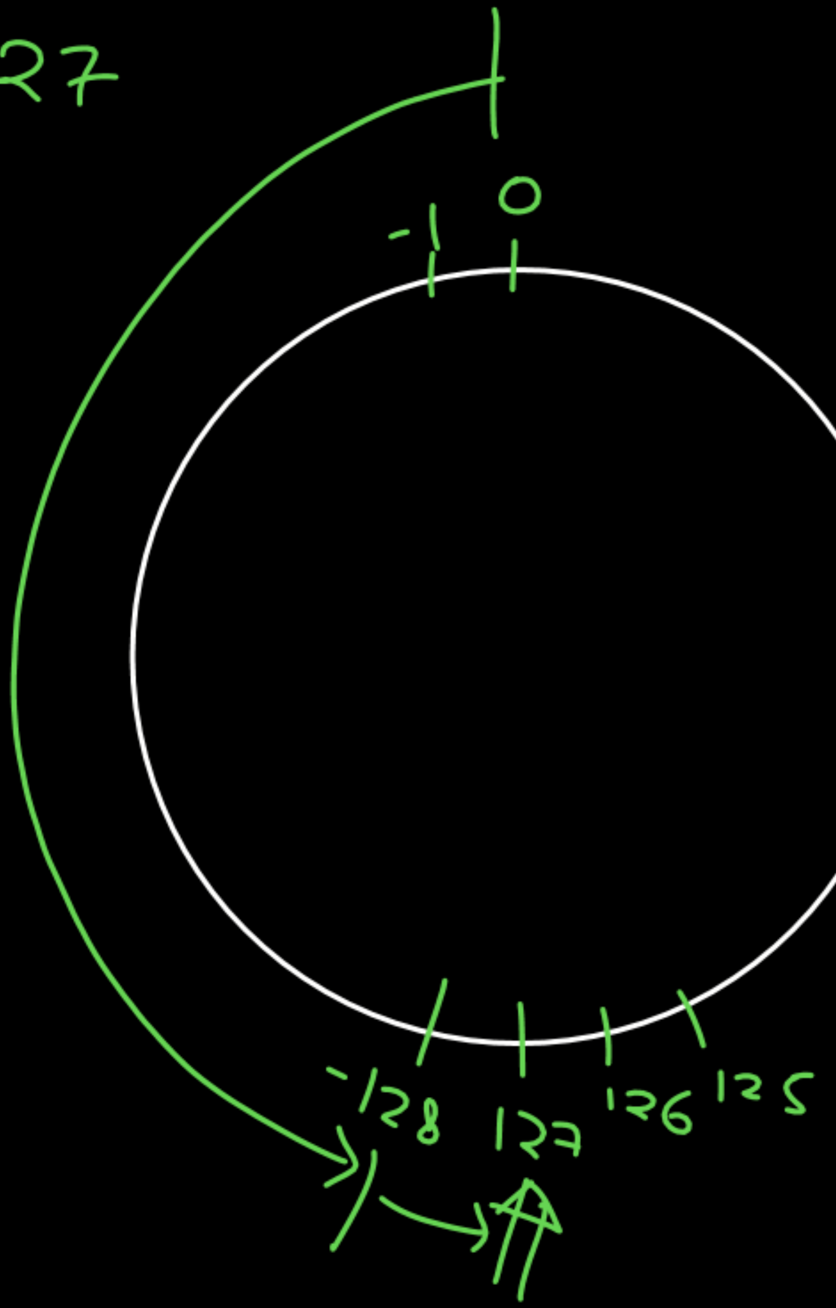


```
char ch = -129 ;

printf("%d",ch); 127
```

ch

$\boxed{127}$



symbol $\Longrightarrow$ 65

```c
char ch = 'A';
```

```c
printf("%d",ch);
```
65

%d → integer val

```c
printf("%c",ch);
```
A

%c → Symbol
char. systm

ch

01000001

65   A

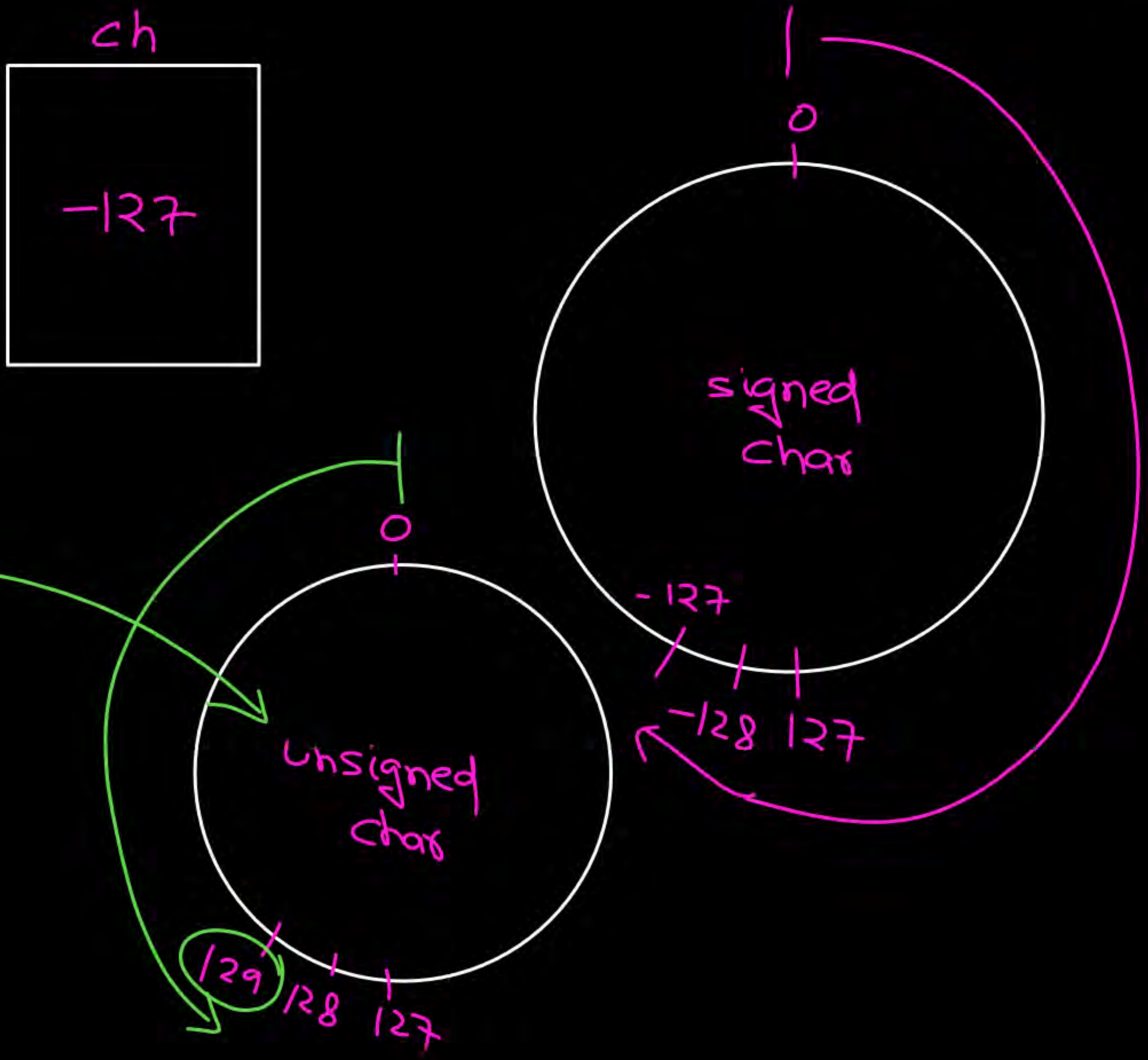
```c
char ch = 129;
printf("%c",ch);
```

char. system

⇓

+ve value

Symbol ASCII Code is 129



& : address of operator

$\boxed{\text{scanf}}$ → (i) format

dest :

A ← 01000001 → 65

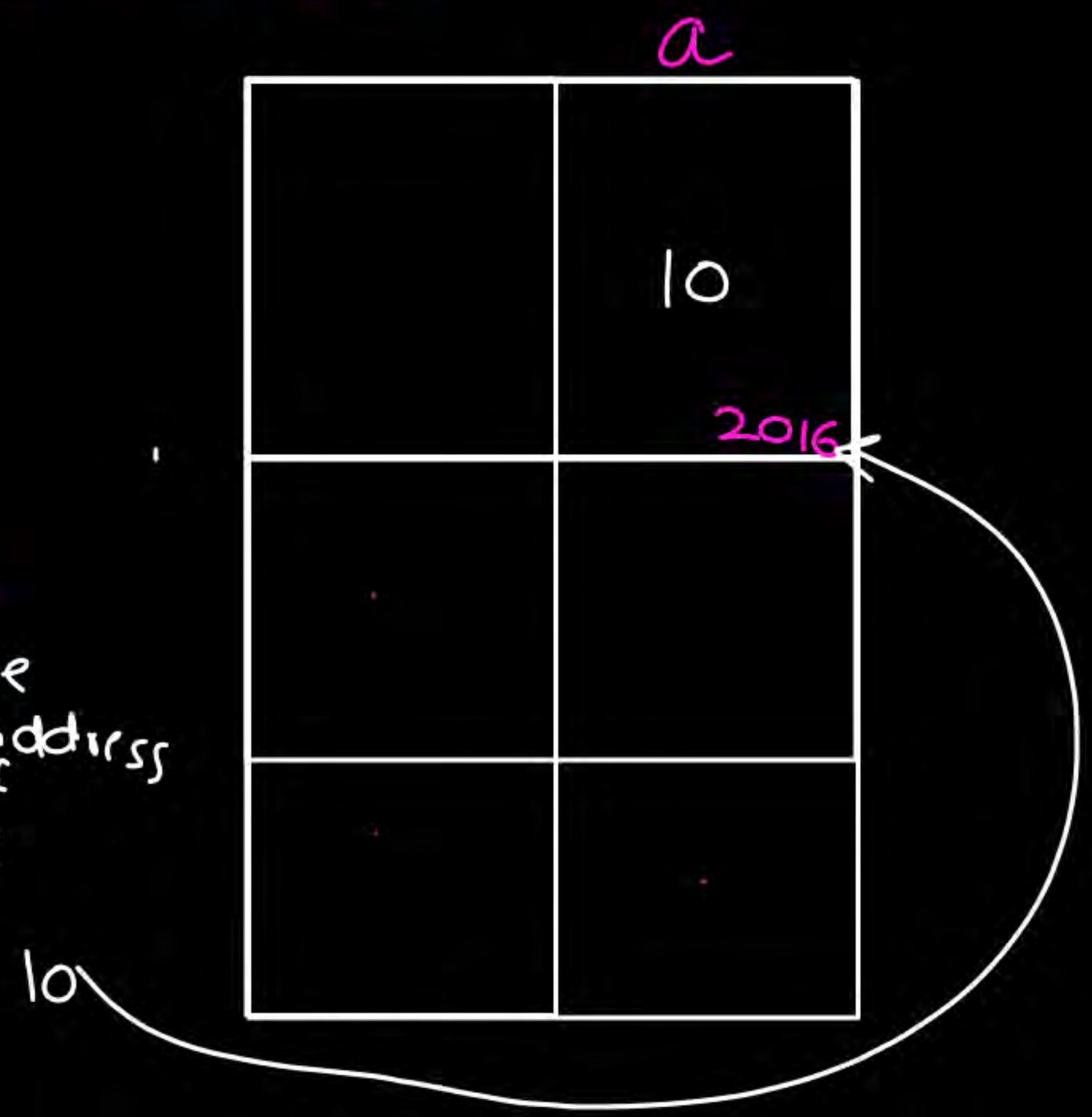
```
int a;

printf("Enter a value");
scanf("%d", &a);
```

read a int. value

store that 10 value at address of a

Enter a value

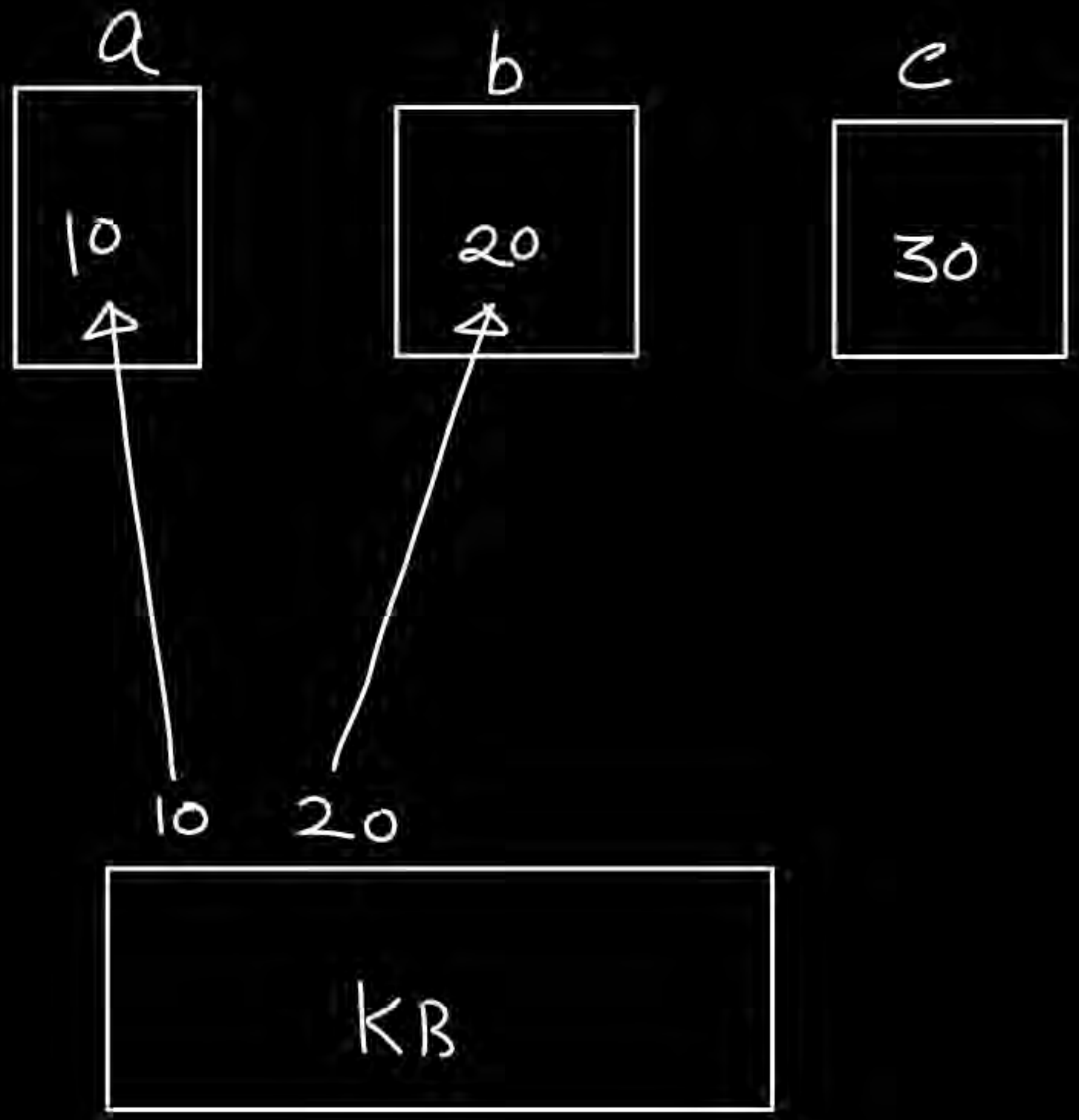


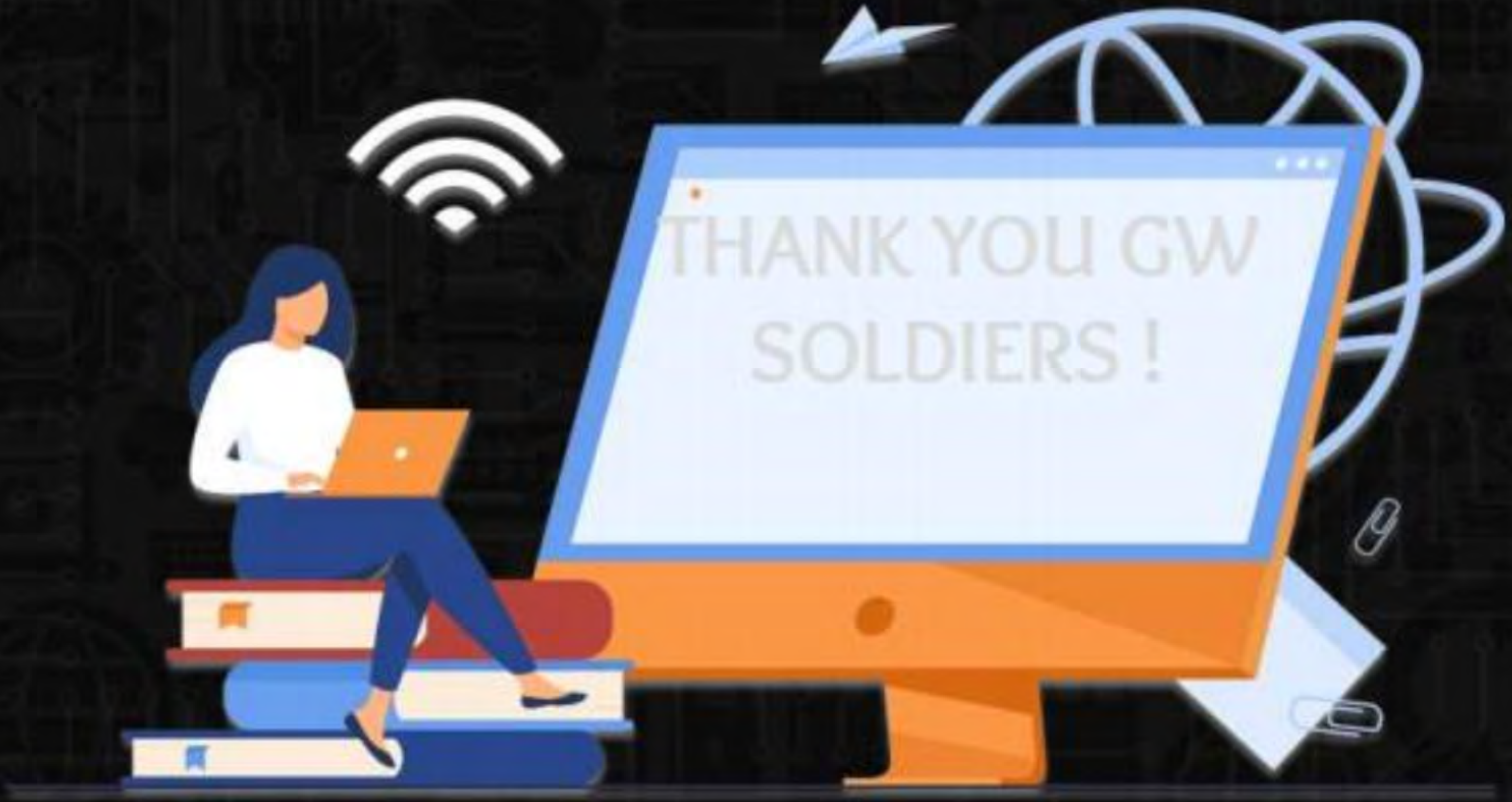
```
int a, b, c;

printf("Enter 2 numbers");
scanf("%d %d", &a, &b);

c = a + b;

printf("The sum is %d", c);
```




THANK YOU GW
SOLDIERS !